路径故障转移
ONTAP 9

NetApp
March 23, 2022

This PDF was generated from https://docs.netapp.com/zh-cn/ontap/concepts/path-failover-concept.html on March 23, 2022. Always check docs.netapp.com for the latest.

目录

路径故障转移 1
路径故障转移概述 1
NAS 路径故障转移 1
SAN 路径故障转移 2

路径故障转移

路径故障转移概述

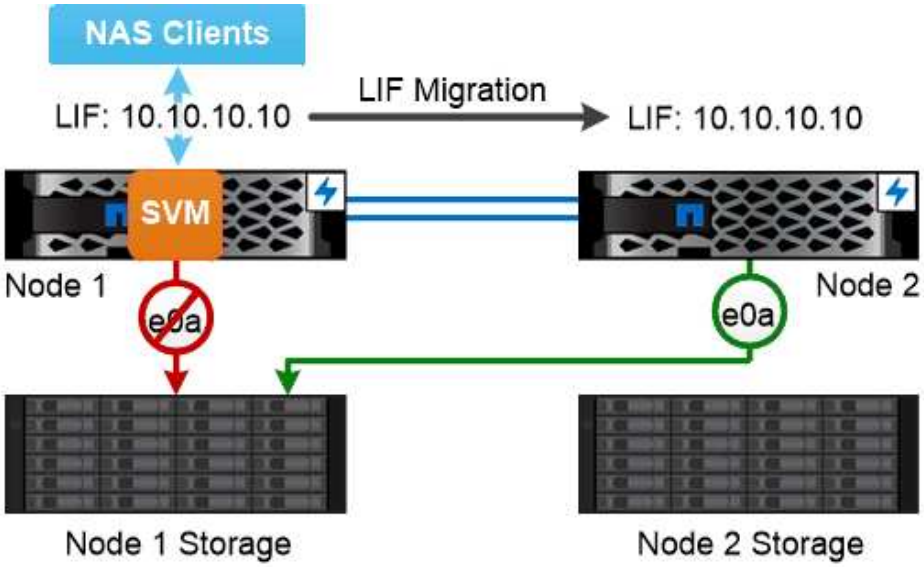
ONTAP 在 NAS 和 SAN 拓扑中管理路径故障转移的方式存在重要差异。链路出现故障后， NAS LIF 会自动迁移到其他网络端口。SAN LIF 不会迁移（除非您在发生故障后手动移动）。相反，主机上的多路径技术会将流量转移到同一 SVM 上的其他 LIF ，但会访问不同的网络端口。

NAS 路径故障转移

在 NAS LIF 的当前端口出现链路故障后，该 LIF 会自动迁移到正常运行的网络端口。LIF 迁移到的端口必须是 LIF 的 *failover group* 的成员。故障转移组策略 _ 可将数据 LIF 的故障转移目标缩小至数据所属节点及其 HA 配对节点上的端口。

为了便于管理， ONTAP 会为网络架构中的每个 *broadcast domain* 创建一个故障转移组。广播域对属于同一第 2 层网络的端口进行分组。例如，如果您使用 VLAN 按部门（工程，营销，财务等）隔离流量，则每个 VLAN 都会定义一个单独的广播域。每次添加或删除广播域端口时，与广播域关联的故障转移组都会自动更新。

几乎始终最好使用广播域定义故障转移组，以确保故障转移组保持最新。但是，有时可能需要定义与广播域不关联的故障转移组。例如，您可能希望 LIF 仅故障转移到广播域中定义的部分端口中的端口。



A NAS LIF automatically migrates to a surviving network port after a link failure on its current port.

- 子网 _*

subnet 会在广播域中保留一个 IP 地址块。这些地址属于同一个第 3 层网络，并在创建 LIF 时分配给广播域中的端口。与指定 IP 地址和网络掩码相比，在定义 LIF 地址时指定子网名称通常更简单，更不容易出错。

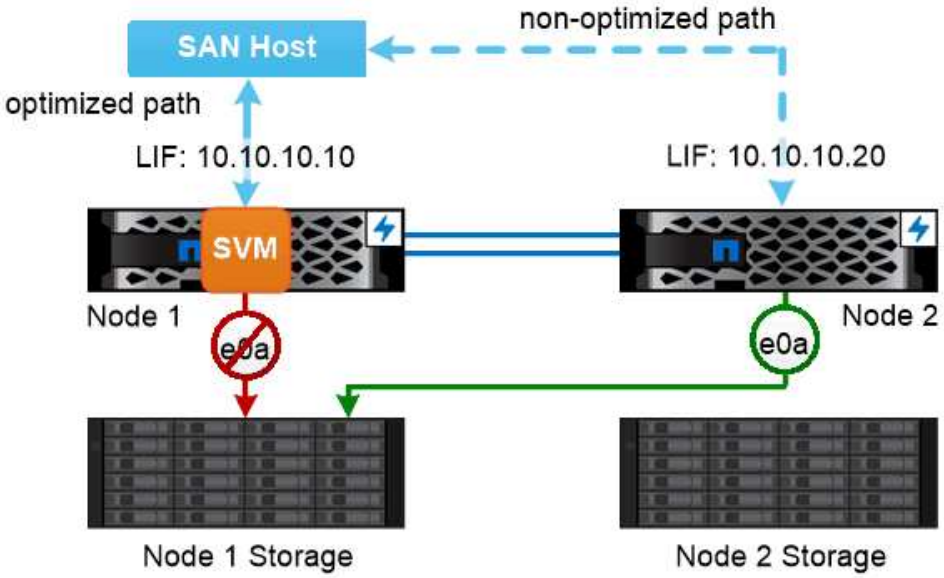
SAN 路径故障转移

在链路出现故障后， SAN 主机使用 ALUA （非对称逻辑单元访问）和 MPIO （多路径 I/O ）将流量重新路由到运行正常的 LIF 。预定义的路径可确定 SVM 所提供的 LUN 的可能路由。

在 SAN 环境中，主机被视为向 lun *targets* 发出请求的 _*initiator* 。 _ MPIO 可启用从启动程序到目标的多个路径。ALUA 标识最直接的路径，称为 _optim优化 的路径。 _

通常，您可以为 LUN 所属节点上的 LIF 配置多个优化路径，并为其 HA 配对节点上的 LIF 配置多个非优化路径。如果所属节点上的一个端口发生故障，则主机会将流量路由到运行正常的端口。如果所有端口都发生故障，主机将通过非优化路径路由流量。

默认情况下， ONTAP 选择性 LUN 映射（ SLM ）会限制从主机到 LUN 的路径数。新创建的 LUN 只能通过 LUN 所属节点或其 HA 配对节点的路径进行访问。您还可以通过在 *port set* 中为启动程序配置 LIF 来限制对 LUN 的访问。



A SAN host uses multipathing technology to reroute traffic to a surviving LIF after a link failure.

在 SAN 环境中移动卷 _*

默认情况下， ONTAP 选择性 LUN 映射（ SLM ） _ 会限制从 SAN 主机到 LUN 的路径数。新创建的 LUN 只能通过 LUN 所属节点或其 HA 配对节点的路径访问，即 LUN 的 *reporting nodes* 。

这意味着，将卷移动到另一个 HA 对上的节点时，您需要将目标 HA 对的报告节点添加到 LUN 映射中。然后，您可以在 MPIO 设置中指定新路径。卷移动完成后，您可以从映射中删除源 HA 对的报告节点。

Copyright Information

Copyright © 2022 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system- without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.